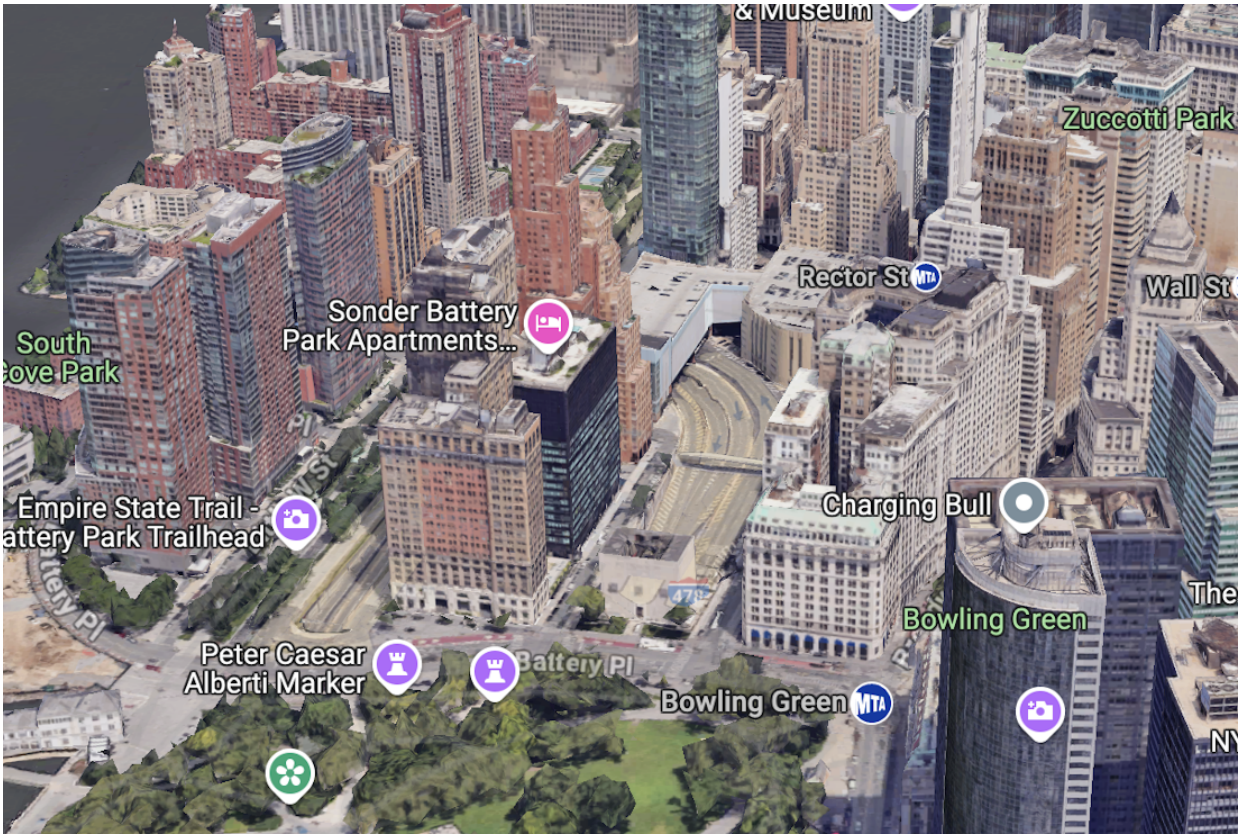


Figure 1.4 Google Maps view of lower Manhattan showing a photorealistic 3D view of the built environment provides a more detailed example of a descriptive digital twin. Imagery © 2025 Google, Map data © 2025, Map data © 2025 Google.

However, the visual representation is only part of the story. Behind the scenes, Google Maps relies on a graph-based model of the road network. Roads are represented as edges in a graph, and intersections are nodes, as shown in figure 1.5. This data structure is what enables turn-by-turn navigation, route optimization, and even traffic prediction. In other words, the visual part of Google Maps is just the surface and underneath is a rich digital model of the physical infrastructure.